

Operating System

CS-2006

Lecture 3

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Traditional

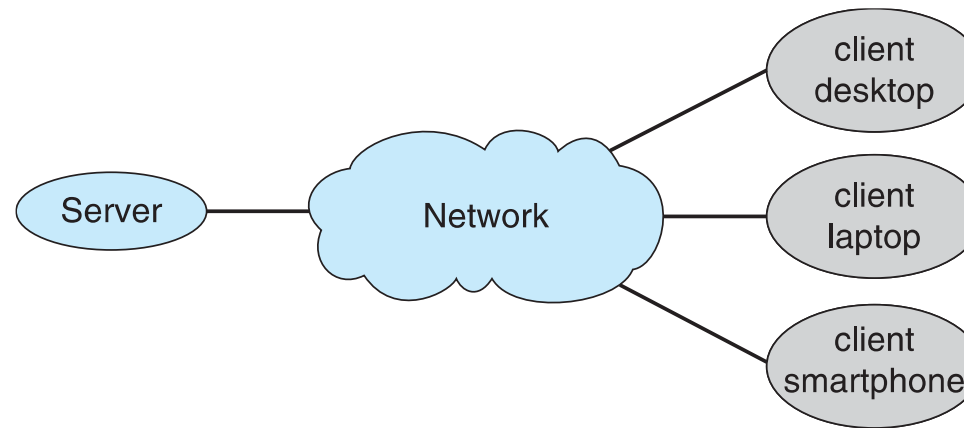
- Stand-alone general-purpose machines
 - computer that is used on its own without requiring a connection to a local area network (LAN)
- Blurred as most systems interconnect with others (i.e., the Internet)
- **Portals provide web access** to internal systems
- Network computers (thin clients) are like Web terminals
- Mobile computers interconnect via wireless networks
- Networking becoming ubiquitous – even home systems use firewalls to protect home computers from Internet attacks

Mobile

- Handheld smartphones, tablets, etc.
- What is the functional difference between them and a “traditional” laptop?
- Extra feature – more OS features (GPS, gyroscope)
- Allows new types of apps like augmented reality
- Use IEEE 802.11 wireless, or cellular data networks for connectivity
- Leaders are Apple iOS and Google Android

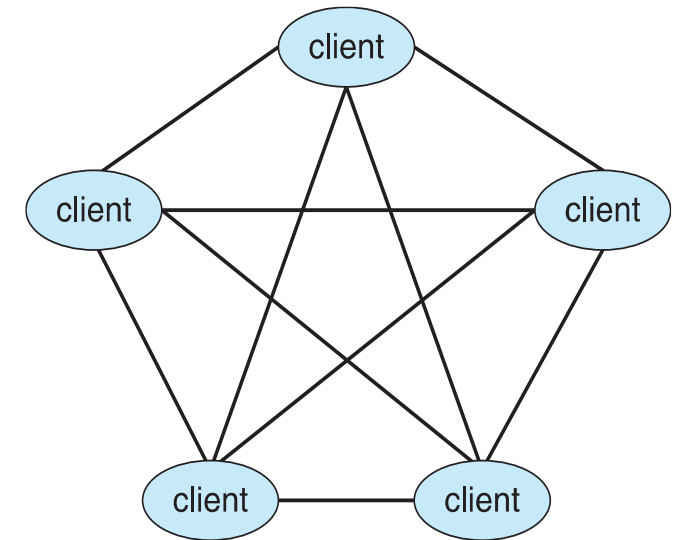
Client Server System

- Client-Server Computing
 - Many systems now servers, responding to requests generated by clients
 - Compute-server system provides an interface to client to request services (i.e., database)
 - File-server system provides interface for clients to store and retrieve files



Peer to Peer System

- Another model of distributed system
- P2P does not distinguish clients and servers
 - Instead all nodes are considered peers
 - May each act as client, server or both
 - Node must join P2P network
 - Registers its service with central lookup service on network, or
 - Broadcast request for service and respond to requests for service via *discovery protocol*
- Examples include **Voice over IP (VoIP)** such as Skype



Cloud Computing

Cloud computing is the delivery of different services through the Internet, **including data storage, servers, databases, networking, and software.**

- Delivers computing, storage, even apps as a service across a network
 - Logical extension of virtualization because it uses virtualization as the base for its functionality.
 - Amazon EC2 has thousands of servers, millions of virtual machines, petabytes of storage available across the Internet, pay based on usage

Cloud Computing (Cont...)

- Three types of cloud computing
 - Private cloud
 - run by a company for the company's own use
 - Example: WMware
 - Public Cloud
 - available via Internet to anyone willing to pay
 - AWS (Amazon Web Services)
 - Hybrid Cloud
 - includes both public and private cloud components
 - Nutanix Cloud Clusters, VMware Cloud Foundation

Cloud Computing Services

Software as a Service (SaaS) (use it)

one or more applications available via the Internet

Example: word processor

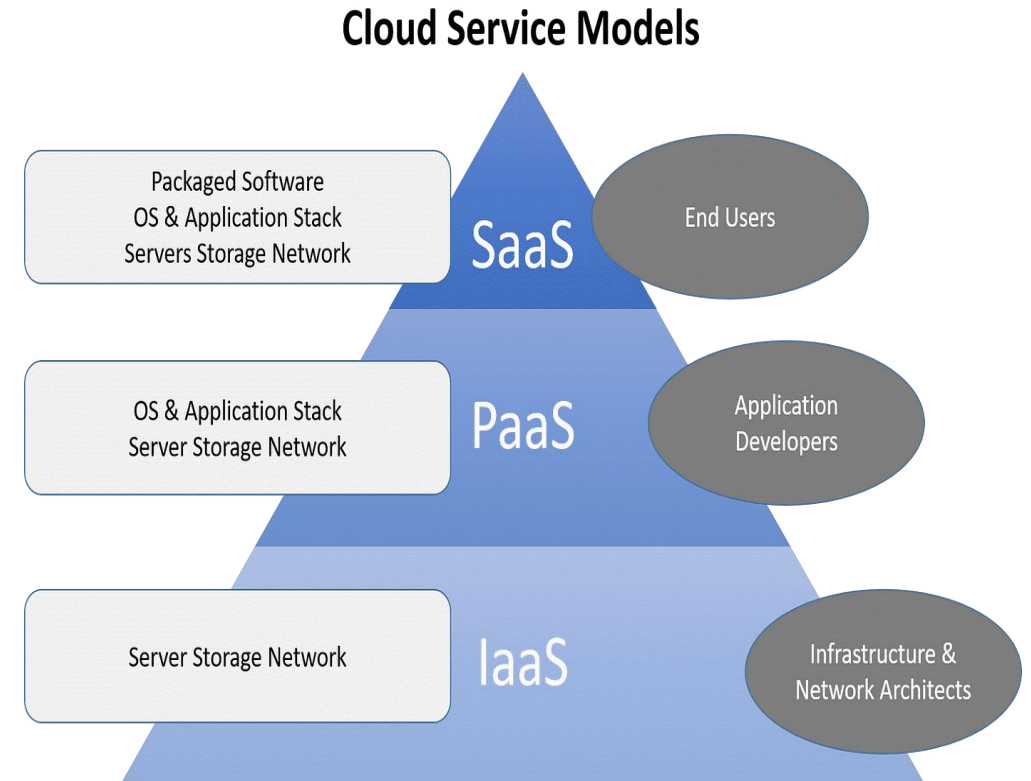
Platform as a Service (PaaS) (Build with it)

software stack ready for application use via the Internet

Example: database servers

Infrastructure as a Service (IaaS) (Move to it)

servers or storage available over Internet
storage available for backup use



Real Time Embedded Systems

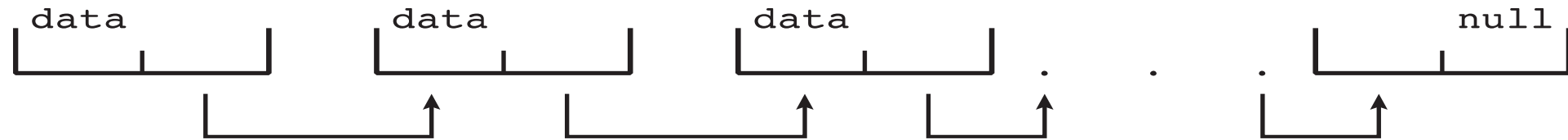
- Real-time embedded systems most prevalent form of computers
 - Vary considerable, special purpose, limited purpose OS, **real-time OS**
 - Use expanding
- Many other special computing environments as well
 - Some have OSes, some perform tasks without an OS
- Real-time OS has well-defined fixed time constraints
 - Processing ***must*** be done within constraint
 - Correct operation only if constraints met

Free and open source Operating System

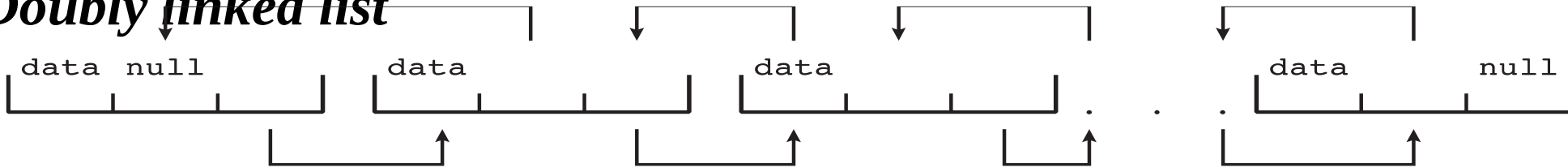
- Operating systems made available in source-code format rather than just binary **closed-source** and **proprietary**
- Counter to the **copy protection** and **Digital Rights Management (DRM)** movement
- Started by **Free Software Foundation (FSF)**, which has “copyleft” **GNU Public License (GPL)**
 - Free software and open-source software are two different ideas championed by different groups of people
 - Examples include **GNU/Linux** and **BSD UNIX** (including core of **Mac OS X**), and many more
- Can use VMM like VMware Player (Free on Windows), Virtualbox (open source and free on many platforms - <http://www.virtualbox.com>)
 - Use to run guest operating systems for exploration

Kernel Data Structures

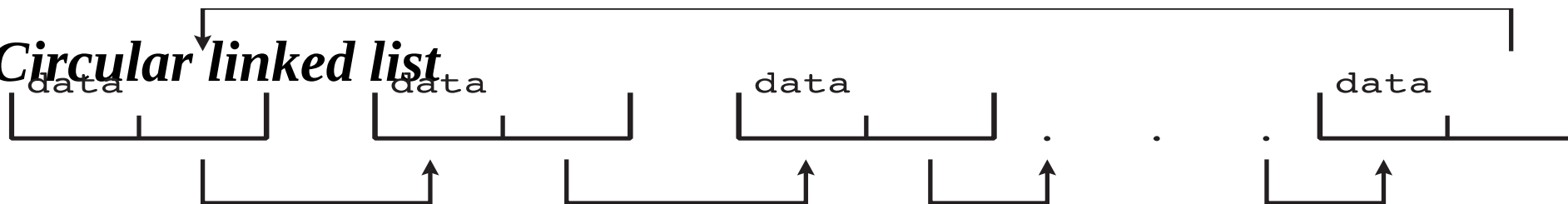
- Many similar to standard programming data structures
- ***Singly linked list***



- ***Doubly linked list***

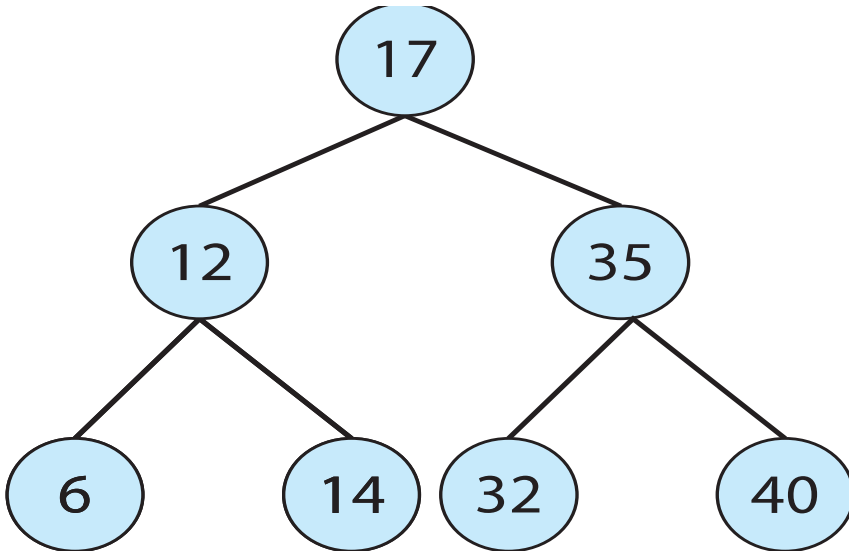


- ***Circular linked list***

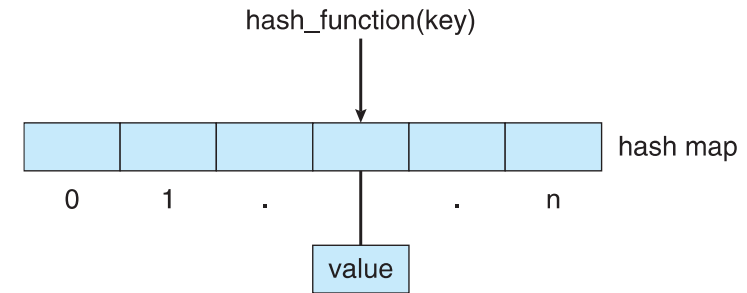


Kernel Data Structures

- **Binary search tree**
left $\leq$ right
 - Search performance is $O(n)$
 - **Balanced binary search tree** is $O(\lg n)$



- **Hash function** can create a **hash map**



- **Bitmap** – string of n binary digits representing the status of n items
- Linux data structures defined in ***include*** files
<linux/list.h>, <linux/kfifo.h>, <linux/rbtree.h>